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Getting prices right on electricity spot markets: On the economic impact of advanced power flow models[☆]

Martin Bichler, Johannes Knörr [*]

School of Computation, Information and Technology, Technical University of Munich, Boltzmannstrasse 3, Garching, 85748, Germany

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ABSTRACT

As the share of variable renewable energy increases, adequate prices on electricity spot markets become increasingly important as they set signals for scarcity, investment, or demand response. Market prices are derived from the underlying welfare maximization problem. On electricity spot markets, this optimization problem is based on the non-convex and non-linear Alternating Current Optimal Power Flow (ACOPF) model. Since the ACOPF is intractable, electricity markets around the world use a linear approximation, the Direct Current Optimal Power Flow (DCOPF) model. Recent research has led to better non-linear relaxations of the ACOPF. We show that these non-linear relaxations increase welfare and imply significantly lower redispatch costs and side-payments. Most importantly, we show that the price signals obtained from non-linear relaxations are much improved. The DCOPF often yields high price differences between nodes when there is no line congestion in the AC-feasible solution or vice versa. Such biased price signals pose a significant problem in practice as they lead to inefficient demand response, distorted investment signals, and incorrect congestion incomes. The use of non-linear relaxations mitigates this problem and provides an important advantage of the resulting prices over prices based on the DCOPF.

1. Introduction

Electricity spot markets collect bids from buyers and sellers and solve a welfare maximization problem to determine the optimal economic dispatch and prices. With an accurate representation of the transmission network and the underlying physics, this can be described as a non-linear and non-convex optimization problem referred to as Alternating Current Optimal Power Flow (ACOPF) problem (Molzahn and Hiskens, 2019). This ACOPF is computationally intractable for the problem sizes that we observe in real-world electricity markets. As a result, linearized network models are used for market clearing and pricing. The standard linearized model is known as Direct Current Optimal Power Flow (DCOPF) problem. The DCOPF or variations thereof are used in most jurisdictions today, but an optimal solution for DCOPF is generally neither AC-optimal nor AC-feasible. This requires market operators or transmission system operators (TSOs) to adjust the dispatch after the market clearing to reach a physically feasible outcome.

This issue is magnified by the transition to renewable energy sources (Lété et al., 2022). A decreasing amount of thermal generators can supply reactive power and the consideration of reactive power on the transmission level is therefore crucial (Hemmati et al., 2013; Karmakar and Bhattacharyya, 2020). Larrahondo et al. (2021) found that high integration of wind power contributes to the inaccuracies of the DCOPF, e.g., by disregarding reactive power. There have been significant efforts to obtain tighter and more accurate relaxations of the ACOPF problem in an effort to leverage recent advances in convex optimization for real-world markets, which culminated in the ARPA-E Grid Optimization Competition.[1] However, errors in the optimization models due to linearization or simplifying assumptions remain a concern in virtually all electricity markets. For example, the revision of the European Capacity Allocation and Congestion Management (CACM) regulation requires accounting for "linearization errors" while calculating available capacities for trading (ACER, 2021). A key concern of linearized models is the welfare loss arising from a poor approximation of ACOPF.

In recent years, substantial research has been devoted to finding tighter convex relaxations and approximations for the ACOPF problem (Molzahn and Hiskens, 2019). This research is driven by advances in convex non-linear optimization algorithms. A goal of the ARPA-E Grid Optimization Challenge is to develop algorithms for the next

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* Corresponding author.
E-mail addresses: bichler@cit.tum.de (M. Bichler), knoerr@cit.tum.de (J. Knörr).

[1] See https://gocompetition.energy.gov/ for further details.

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generation of solvers providing tighter relaxations of ACOPF and to better reflect the physics of the power grid. This research focuses on the optimality of the solution and computational costs to obtain it.

However, even if one could solve ACOPF to optimality, getting adequate price signals is challenging. Competitive equilibrium (aka. Walrasian) prices in economic theory leverage duality theory in convex optimization. Walrasian prices in a convex market are such that no participant would want to deviate from the efficient outcome (aka. envy-freeness), and the market is budget-balanced. This means that market operators neither need to subsidize the market, nor do they make a profit. Unfortunately, a well-known problem in economics is that Walrasian prices generally exist only when the allocation problem is a convex optimization problem (Bichler et al., 2020). Many real-world markets are based on non-convex optimization problems with electricity spot markets as a prime example. Such non-convexities can even affect transmission network expansion planning (García-Cerezo et al., 2021)

The literature on electricity market pricing developed a number of proposals how to price electricity in the presence of such non-convexities (Liberopoulos and Andrianesis, 2016). The most prominent examples are Integer Programming (IP) pricing (O'Neill et al., 2005) and Convex Hull (CH) pricing (Gribik et al., 2007; Hogan and Ring, 2003; Bichler et al., 2023). IP pricing is used in a number of U.S. markets and so are versions of CH pricing. Both pricing rules require side-payments by the market operator to compensate losses of market participants (make-whole payments, MWPs) or even to compensate all incentives to deviate from the efficient allocation (lost opportunity costs, LOCs, for buyers and sellers; potential congestion revenue shortfalls, PCRS, for TSOs). However, the high MWPs are a concern of the U.S. Federal Energy Regulatory Commission (FERC).[2] The U.S. FERC also found that the size and concentration of side-payments can be affected by the inability to accurately model ACOPF, suggesting that the benefit-to-cost ratio of more accurate power flow representations is at least 100-fold (Cain and O'Neill, 2012).

We focus on the improvement of price signals we can expect from better relaxations of ACOPF. Ideally, prices reflect marginal costs at a network node to provide adequate incentives for investment or demand reduction to aid overall system stability (Herrero et al., 2015). If DCOPF does not adequately reflect the physics of the power grid, the prices retrieved from the DCOPF solution will not be accurate. Thus, tighter relaxations of ACOPF could lead to prices that better reflect scarcity in the physical network. However, the magnitude of these improvements is unclear. So far, only very few articles study pricing based on non-linear convex relaxations of ACOPF (Ndrio et al., 2019; Papavasiliou, 2018; Garcia et al., 2020; Winnicki et al., 2020). All of these papers focus on small networks with a few nodes and they assume convex cost functions of sellers and fixed demand of buyers. Non-convexities of sellers' cost or buyers' valuation functions are important and have been the central problem that sparked the literature on electricity market pricing (Liberopoulos and Andrianesis, 2016; Zoltowska, 2016; Kuang et al., 2019).

We study the impact of non-linear convex market clearing models on prices, MWPs, LOCs, and PCRS. In contrast to prior work, we consider non-convexities in the preferences of buyers and sellers and analyze realistic problem sizes, which should provide a good estimate of the prices and welfare gains we can expect from tighter power flow relaxations in the field. More precisely, we consider the linearized DCOPF, a second-order conic (SOC) relaxation, and a quadratic convex (QC) relaxation (Molzahn and Hiskens, 2019).[3] For each of these relaxations, we compute Integer Programming (IP) prices (O'Neill et al., 2005) and Convex Hull (CH) prices (Gribik et al., 2007; MISO, 2023; Hua and Baldick, 2017) as the two pricing rules primarily used in U.S. markets.[4] To our knowledge, this is the first paper to simultaneously consider non-convexities arising from power flow models as well as non-convexities arising from bidding formats. We leverage the large problem instances from the ARPA-E Grid Challenge II[5] as they were designed to reflect real-world problems.

The results of our extensive numerical experiments show that different power flow models lead to substantial differences in the allocation. This of course has an impact on both welfare and prices. If the dispatch computed on the market is not AC-feasible, redispatch is required. Therefore, we report the costs of redispatch as well as welfare, MWPs, LOCs, and PCRS after computing a near AC-feasible solution.

As one would expect, tighter convex relaxations require substantially less redispatch compared to the standard DCOPF approximation, and MWPs, LOCs and PCRS are, on average, lower for the final AC-feasible outcome. Moreover, the tighter convex relaxations lead to higher welfare of the final dispatch. However, welfare gains are not the most important argument for tighter convex relaxations of ACOPF. Our numerical experiments show that the prices obtained from DCOPF are often substantially different from those of the non-linear relaxations, leading to biased scarcity signals that distort effective demand response and investment decisions. In contrast, the results of the SOC and QC relaxation, which both model reactive power and line losses more accurately, are almost identical. Prices obtained from DCOPF might be excessively high at some of the nodes, even though there is no congestion in the AC-feasible solution. In other words, DCOPF leads to unnecessary price peaks at some of the nodes, with prices being multiples of the average price, even though there is no congestion in the physical grid at all. Similarly, we might encounter congestion in the AC-feasible solution that the DC prices of adjacent nodes do not reflect. Overall, DCOPF can expose some of the market participants to unjustifiably high prices and further leads to inefficient demand response, biased investment signals, and wrong congestion incomes. In a world with 100% renewables, where adequate demand response is even more important, this is a decisive disadvantage of standard DCOPF solutions compared to tighter non-linear relaxations.

2. Power flow models and electricity pricing

In this section, we describe the setup of the electricity market we consider. We outline the optimization problem to be solved by the market operator and discuss competitive equilibrium theory with respect to pricing on non-convex markets.

2.1. Setup

We consider a coupled electricity market with a set of buyers (consumers) B and a set of sellers (generators) S of electricity located at interlinked nodes N in an electricity grid.[6] The set of power lines L is encoded by the pairs of nodes (i,k) that are directly connected by lines. We consider a single period such that the set of goods corresponds to units of (real or reactive) power at the locations N. Each buyer $b \in B$ possesses a valuation function $v_b : \mathbb{C}^{|N|} \to \mathbb{R}$ and each seller $s \in S$ possesses a cost function $c_s : \mathbb{C}^{|N|} \to \mathbb{R}$ that encodes feasibility constraints and preferences.

A market operator collects buy and sell bids and provides a feasible allocation (x, y) as well as locational prices λ. The variables $x \in \mathbb{C}^{|N|\times|B|}$

[2] See https://www.ferc.gov/industries-data/electric/electric-power-markets/energy-price-formation for further details.

[3] We also considered a semidefinite programming (SDP) relaxation, but it was not possible to solve large mixed-integer SDPs to optimality for the problem sizes at hand.

[4] CH pricing is currently used for fast-start pricing by MISO and ISO-NE (MISO, 2023; PJM, 2018).

[5] See https://gocompetition.energy.gov/challenges/challenge-2 for further details and to access the data.

[6] One may aggregate several nodes into zones as it is done in European markets, but we assume a fully nodal setting.

(with column vectors $x_b \in \mathbb{C}^{|N|}$) and $y \in \mathbb{C}^{|N|\times|S|}$ (with column vectors $y_s \in \mathbb{C}^{|N|}$) describe the allocated bundles to buyers and sellers, respectively. In essence, the market operator is faced with the following welfare maximization problem:

$$\max_{x,y} \quad \sum_{b\in B} v_b(x_b) - \sum_{s\in S} c_s(y_s) \tag{1}$$

subject to $\quad x, y \in \Psi.$

Here, Ψ describes the set of feasible power flows. In other words, the market operator seeks the welfare-maximizing allocation that ensures feasible power flows with strict supply–demand equivalence at each node. However, with an accurate representation of the underlying physics, Ψ is highly non-linear and non-convex, and solving such problems to optimality is not viable from a practical perspective. Therefore, various relaxations and approximations have been introduced that reduce Ψ to a convex set. We discuss possible power flow models in Section 2.2.

Given an optimal solution (x^*, y^*), the market operator proceeds by determining adequate prices $\lambda \in \mathbb{R}^{|N|}$ (we assume non-negative prices for real power and prices of zero for reactive power). However, by design of the bidding formats, valuation functions v_b and cost functions c_s are typically non-convex. In this case, Walrasian equilibrium prices do not exist and market operators need to resort to non-uniform prices and consider incentives to deviate. We provide an overview of these issues in Section 2.3. It should be noted that some market operators (e.g., in European markets) deviate from the welfare-maximizing allocation in order to ensure better price properties. Since this comes with computational challenges (All NEMO Committee, 2022), we focus on pricing optimal outcomes in this paper.

Both the complexity of power flows and of pricing in non-convex markets has been studied extensively, but typically separated. Practical electricity markets employ fully linearized versions of Ψ and pricing rules that require side-payments to some market participants. The interdependence between alternative power flow models, different pricing rules, and the welfare of the resulting outcome has not yet been investigated.

2.2. Power flow

In essence, power flow models describe the relationship between power injections/withdrawals at each node and the resulting line flows and voltage phasors. Due to their complexity and fundamental importance for power systems, there is a huge strand of literature dealing with power flow problems in various forms and aspects. We refer to Cain and O'Neill (2012), Frank et al. (2012) Castillo and O'Neill (2013), Low (2014a) and Molzahn and Hiskens (2019) for comprehensive overviews.

Modeling the physics of a power grid results in a highly non-linear and non-convex optimization problem, the Alternating Current Optimal Power Flow (ACOPF) problem. As the intention of this paper is to connect the literature on power models and pricing with non-convexities, we will neither discuss the physics of electric circuits that lead to these types of constraints, nor the numerous representations of ACOPF that have been proposed. We refer interested readers to Bienstock et al. (2020) and Molzahn and Hiskens (2019). For our purposes, we provide a bus injection formulation of ACOPF (Molzahn and Hiskens, 2019) below. We use $\mathrm{Re}(x_b)$ to denote the real power consumed by buyer b and $\mathrm{Im}(x_b)$ to denote the reactive power (and analogously for sellers s). Parameters that are given include a complex nodal admittance matrix $Y \in \mathbb{C}^{|N|\times|N|}$ with $Y = G + jB$ consisting of a real conductance part G and an imaginary susceptance part B, as well as minimum/maximum voltages V_i^{min}, V_i^{max} at each node $i \in N$. With $V_i = V_{di} + jV_{qi}$ as complex voltage phasor variable, the set of feasible power flows Ψ^{AC} can be written as

$$\begin{aligned}
\Psi^{AC} = \{ & x \in \mathbb{C}^{|N|\times|B|}, y \in \mathbb{C}^{|N|\times|S|} : \\
& \sum_{s\in S} \mathrm{Re}(y_{is}) - \sum_{b\in B} \mathrm{Re}(x_{ib}) = \sum_{k\in N} V_{di}(G_{ik}V_{dk} - B_{ik}V_{qk}) \\
& \quad + V_{qi}(B_{ik}V_{dk} + G_{ik}V_{qk}) \quad \forall i \in N, \\
& \sum_{s\in S} \mathrm{Im}(y_{is}) - \sum_{b\in B} \mathrm{Im}(x_{ib}) = \sum_{k\in N} V_{di}(-B_{ik}V_{dk} - G_{ik}V_{qk}) \\
& \quad + V_{qi}(G_{ik}V_{dk} - B_{ik}V_{qk}) \quad \forall i \in N, \\
& V_{di}^2 + V_{qi}^2 = |V_i|^2 \quad \forall i \in N, \\
& (V_i^{min})^2 \le |V_i|^2 \le (V_i^{max})^2 \quad \forall i \in N, \\
& V_i \in \mathbb{C} \quad \forall i \in N \quad \}.
\end{aligned}$$

The set of constraints may further include limits on line flows or the definition of a reference bus. In any case, the quadratic and non-convex constraints implied by this system of equations make the ACOPF problem notoriously hard to solve and it is considered intractable in spite of advances in global optimization. The desire to obtain good solutions to ACOPF has sparked competitions such as the ARPA-E Grid Optimization Competition.[7] In this context, new reformulations and decomposition techniques have been proposed (Petra and Aravena, 2021).

Overall, there has been significant research on convex relaxations and approximations of ACOPF to provide tractable programs and high quality solutions for practical application. These relaxations make use of advances in convex optimization, i.e. they leverage increasingly powerful solvers for linear programs (LPs), second-order cone programs (SOCPs), and semidefinite programs (SDPs). We briefly outline the main ideas of the relaxations used in this paper and refer to more comprehensive summaries (Zohrizadeh et al., 2020; Molzahn and Hiskens, 2019) for further details. Molzahn and Hiskens (2016) provide an accessible illustrative example of different convex relaxations.

SDP relaxation

The SDP-relaxation, dating back to the works of Shor (1987), is based on lifting decision variables of the canonical ACOPF to capture non-convexities in a single constraint. In particular, defining $W = VV^T$ as the product of voltages, we can rewrite the quadratic terms in Ψ^{AC} in linear terms of entries of W. Constraining W to be a positive semidefinite and rank-one matrix, i.e. $W \succcurlyeq 0$ and $\mathrm{rank}(W) = 1$, is a sufficient condition for the exactness of this reformulation. Since the rank-one constraint now contains all non-convexities resulting from power flows, dropping this constraint yields a convex SDP relaxation.

With $\langle \cdot, \cdot \rangle$ as Frobenius inner product and e_i as ith unit vector, the relaxed set of feasible power flows Ψ^{SDP} can be written as (Bai et al., 2008; Molzahn and Hiskens, 2019)

$$\begin{aligned}
\Psi^{SDP} = \{ & x \in \mathbb{C}^{|N|\times|B|}, y \in \mathbb{C}^{|N|\times|S|} : \\
& \sum_{s\in S} \mathrm{Re}(y_{is}) - \sum_{b\in B} \mathrm{Re}(x_{ib}) = \langle L_{P_i}, W \rangle \quad \forall i \in N, \\
& \sum_{s\in S} \mathrm{Im}(y_{is}) - \sum_{b\in B} \mathrm{Im}(x_{ib}) = \langle L_{Q_i}, W \rangle \quad \forall i \in N, \\
& \quad |V_i|^2 = \langle M_i, W \rangle \quad \forall i \in N, \\
& \quad W \succcurlyeq 0 \quad \}.
\end{aligned}$$

Here, the matrices L_{P_i}, L_{Q_i}, and M_i are defined as

$$L_{P_i} = \frac{1}{2}\begin{pmatrix} \mathrm{Re}(Y^T e_i e_i^T + e_i e_i^T Y) & \mathrm{Im}(Y^T e_i e_i^T - e_i e_i^T Y) \\ \mathrm{Im}(e_i e_i^T Y - Y^T e_i e_i^T) & \mathrm{Re}(Y^T e_i e_i^T + e_i e_i^T Y) \end{pmatrix}$$

$$L_{Q_i} = -\frac{1}{2}\begin{pmatrix} \mathrm{Im}(Y^T e_i e_i^T + e_i e_i^T Y) & \mathrm{Re}(e_i e_i^T Y - Y^T e_i e_i^T) \\ \mathrm{Re}(Y^T e_i e_i^T - e_i e_i^T Y) & \mathrm{Im}(Y^T e_i e_i^T + e_i e_i^T Y) \end{pmatrix}$$

$$M_i = \begin{pmatrix} e_i e_i^T & 0 \\ 0 & e_i e_i^T \end{pmatrix}.$$

[7] See https://gocompetition.energy.gov/ for further details.

While SDP solvers are not as mature as classical LP software, many commercial packages can approximately solve these continuous SDPs in polynomial time using interior point algorithms (Ben-Tal and Nemirovski, 2001; Vandenberghe and Boyd, 1996). Exploiting the sparsity of the network (and thus of matrix W) can further alleviate computational challenges, i.e. by using the matrix completion theorem to impose semidefiniteness constraints only on smaller submatrices of W (Grone et al., 1984).

The SDP relaxation was much celebrated due to its capability to solve several IEEE test cases exactly (Lavaei and Low, 2012).[8] However, the relaxation fails to be exact in general and exactness can even depend on the specific formulation of the problem (i.e. the exactness does not merely depend on physical characteristics of the network) (Kocuk et al., 2016a; Bukhsh et al., 2013; Lesieutre et al., 2011; Molzahn and Hiskens, 2019; Low, 2014b). As a consequence, there are efforts to further strengthen the relaxation, e.g., using the Lasserre hierarchy (Lasserre, 2001, 2009; Josz and Henrion, 2016; Molzahn and Hiskens, 2014, 2015), bound tightening, or lifted non-linear cuts (Coffrin et al., 2015). However, the computational hardness of the problem grows quickly in this case. It should further be noted that mixed-integer SDP solvers, which become relevant when we consider non-convexities in the valuation and cost functions, are still in early stages of development (Gally et al., 2018) and cannot be reliably used for global optimization.[9] Therefore, we could not obtain meaningful results from the SDP relaxation and refrain from reporting approximate results that have little significance.

SOC relaxation

In contrast to SDPs, second-order cone programs (SOCPs) can be efficiently solved by a variety of solvers, even in a mixed-integer form, using interior point methods (Andersen et al., 2003; Ben-Tal and Nemirovski, 2001; Alizadeh and Goldfarb, 2003). However, since SDPs represent a generalization of SOCPs, SOCP relaxations are usually less tight. In this paper, we consider two second-order cone relaxations of ACOPF: Jabr's second-order cone (SOC) relaxation and Hijazi's quadratic convex (QC) relaxation.

Jabr's SOC relaxation (Jabr, 2006; Molzahn and Hiskens, 2019) is similar to the SDP relaxation in that first lifted decision variables are defined from squared voltage magnitudes and the product of voltage phasors, i.e. $c_i = V_{di}^2 + V_{qi}^2$, $cr_{ik} = V_{di}V_{dk} + V_{qi}V_{qk}$, and $ci_{ik} = V_{di}V_{qk} - V_{qi}V_{dk}$. Then, in a radial network, which was the original setting considered by Jabr (2006), an exact representation of Ψ^{AC} can be formulated using these variables.[10] Similar to the SDP relaxation, all non-convexities are contained in a single quadratic equality constraint $cr_{ik}^2 + ci_{ik}^2 = c_i c_k$. Replacing this by an inequality constraint yields a convex SOC relaxation, i.e.,

$$
\begin{aligned}
\Psi^{SOC} = \{ & x \in \mathbb{C}^{|N|\times|B|}, y \in \mathbb{C}^{|N|\times|S|} : \\
& \sum_{s\in S} \mathrm{Re}(y_{is}) - \sum_{b\in B} \mathrm{Re}(x_{ib}) = G_{ii}c_i + \sum_{k\in N\setminus\{i\}} G_{ik}cr_{ik} - B_{ik}ci_{ik} \quad \forall i \in N, \\
& \sum_{s\in S} \mathrm{Im}(y_{is}) - \sum_{b\in B} \mathrm{Im}(x_{ib}) = -B_{ii}c_i + \sum_{k\in N\setminus\{i\}} -B_{ik}cr_{ik} - G_{ik}ci_{ik} \quad \forall i \in N, \\
& cr_{ik} = cr_{ki} \quad \forall (i,k) \in L, \\
& ci_{ik} = -ci_{ki} \quad \forall (i,k) \in L, \\
& cr_{ik}^2 + ci_{ik}^2 \le c_i c_k \quad \forall (i,k) \in L, \\
& c_i, cr_{ik}, ci_{ik} \in \mathbb{R} \quad \forall (i,k) \in L \quad \}.
\end{aligned}
$$

This SOC relaxation can also be obtained from the SDP relaxation by replacing the positive semidefiniteness constraint of W by second-order conic constraints on 2×2 submatrices thereof. As this constitutes a necessary but not sufficient condition for $W \succcurlyeq 0$, the SDP relaxation is at least as tight as the SOC relaxation. Therefore, the SOC relaxation is exact only under restrictive conditions (Jabr, 2006; Gan et al., 2012; Low, 2014b), and tightening techniques can be used to improve the quality of the solution (Kocuk et al., 2016b).

QC relaxation

Hijazi's QC relaxation (Hijazi et al., 2017; Coffrin et al., 2016) is derived from a polar representation of ACOPF. In essence, it uses convex envelopes for various non-convex terms to yield a second-order cone program. By accounting for voltage angle constraints in loops, it is explicitly designed to be applicable to mesh networks (Molzahn and Hiskens, 2019), unlike the SOC relaxation. In fact, the QC relaxation simply tightens Ψ^{SOC} by adding more constraints on voltage angles and magnitudes. We will write $\langle\cdot\rangle$ to denote convex envelopes of non-convex square, bilinear, sine, or cosine terms and refer to Hijazi et al. (2017) and Molzahn and Hiskens (2019) for explicit formulations of these convex envelopes. With this notation and θ_i as voltage angle at node i, the set of power flows can be written as

$$
\begin{aligned}
\Psi^{QC} = \Psi^{SOC} \cap \{ & x \in \mathbb{C}^{|N|\times|B|}, y \in \mathbb{C}^{|N|\times|S|} : \\
& c_i \in \langle |V_i|^2 \rangle \quad \forall i \in N, \\
& cr_{ik} \in \langle \langle |V_i||V_k| \rangle \langle \cos(\theta_i - \theta_k) \rangle \rangle \quad \forall (i,k) \in L, \\
& ci_{ik} \in \langle \langle |V_i||V_k| \rangle \langle \sin(\theta_i - \theta_k) \rangle \rangle \quad \forall (i,k) \in L, \\
& c_i, cr_{ik}, ci_{ik}, \theta_i \in \mathbb{R} \quad \forall (i,k) \in L \quad \}.
\end{aligned}
$$

Note that phase angle differences $\theta_i - \theta_k$ are assumed to be bound between zero and $\frac{\pi}{2}$ (Coffrin et al., 2016). By design, the QC-relaxation is at least as tight as the SOC relaxation while still maintaining computational efficiency as a second-order cone program. Neither the SDP nor the QC relaxation dominate the other.

DC approximation

Finally, we consider the standard linear approximation for ACOPF as it is currently applied in practical electricity markets. The Direct Current Optimal Power Flow (DCOPF) problem (Stott et al., 2009; Li and Bo, 2007; Overbye et al., 2004; Molzahn and Hiskens, 2019; Eldridge et al., 2018) makes three simplifying assumptions: (i) line resistances and reactive power are ignored, (ii) voltage magnitudes $|V_i|$ at each bus are set to 1, and (iii) voltage angle differences between nodes are assumed to be small such that $\sin(\theta_i - \theta_k) \approx \theta_i - \theta_k$ and $\cos(\theta_i - \theta_k) \approx 1$. As a result, the set of power flows simply formulates as a set of linear constraints, i.e.,

$$
\begin{aligned}
\Psi^{DC} = \{ & x \in \mathbb{C}^{|N|\times|B|}, y \in \mathbb{C}^{|N|\times|S|} : \\
& \sum_{s\in S} \mathrm{Re}(y_{is}) - \sum_{b\in B} \mathrm{Re}(x_{ib}) = \sum_{k\in N:(i,k)\in L} -B_{ik}(\theta_i - \theta_k) \\
& \quad - \sum_{k\in N:(k,i)\in L} -B_{ki}(\theta_k - \theta_i) \quad \forall i \in N, \\
& \theta_i \in \mathbb{R} \quad \forall i \in N \quad \}.
\end{aligned}
$$

Of course one may impose an admissible range on voltage angles, e.g. $\theta_i \in [-\frac{\pi}{2}, \frac{\pi}{2}]$. We also note that many system operators in practice use refined linear approximations that go beyond the standard DCOPF (Li et al., 2022).[11] In any case, the set of constraints constitute a polyhedron, and mature and powerful linear programming solvers can be applied (Bertsimas and Tsitsiklis, 1997). At the same time, Ψ^{DC} may not approximate Ψ^{AC} sufficiently well. While there are ambiguous empirical findings on the accuracy of DCOPF (Molzahn and Hiskens, 2019), recent developments in research and practice alike suggest that tighter relaxations would be preferable as long as tractability can be ensured. This motivates to examine the introduced non-linear network models in the context of pricing on wholesale electricity markets.

[8] Note that for the SDP relaxation the rank-one condition of W provides a simple way to verify exactness.

[9] See http://www.opt.tu-darmstadt.de/scipsdp/ for an example of a solver.

[10] Note that for mesh networks, this representation constitutes a relaxation as voltage angles in a loop may not sum to zero (Molzahn and Hiskens, 2019).

[11] For example, approximation models in practice include more constraints that reflect specific knowledge of the system. We therefore remark that practical implementations of DCOPF are likely to be tighter than Ψ^{DC} presented here and may yield better numerical results.

Redispatch

As discussed, the introduced relaxations and approximations are generally not tight, i.e., $\Psi^{AC} \subseteq \Psi^{SDP} \subseteq \Psi^{SOC}$, $\Psi^{AC} \subseteq \Psi^{QC}$ and for Ψ^{DC} we cannot make any statement. Solutions obtained from these power flow models may thus not be physically feasible. It is therefore necessary to reconstruct a feasible solution from Ψ^{AC} given the optimal solution to the relaxation/approximation. As noted by Petra and Aravena (2021), this reconstruction may be as complex as finding a solution from Ψ^{AC} from scratch. Ensuring AC-feasibility often involves iterative processes with metaheuristics which are ineffective when the system is stressed (Castillo, 2016). Thus, the economic costs of reconstructing an AC-feasible solution is another important factor to consider. We will refer to this reconstruction as *redispatch*.[12]

For our purposes, we will consider a cost-based redispatch. When the market operator detects that the obtained solution is not in Ψ^{AC}, the dispatch of certain market participants will be adjusted upward or downward to ensure AC-feasibility. The affected market participants are compensated for their additional costs or lost profits and are thus indifferent to the initially assigned dispatch. We refer to the sum of these compensation payments as *redispatch costs*. The redispatch costs serve as additional benchmarks for efficiency losses of the different relaxations and approximations.

In order to find a near AC-feasible solution, we construct an AC power flow model $x, y \in \Psi^{AC}$, albeit with no objective function. In order to speed up the calculations, we use a warm-start procedure and set starting values for various variables. In particular, the voltage magnitudes, active and reactive power generation, as well as active and reactive power consumption of the solution of the relaxation/approximation are provided. Moreover, we add constraints that restrict the active power generation and consumption to only deviate by some specified tolerance from the initial solution. If the tolerance is chosen too restrictive, the problem may become infeasible, yet an appropriately chosen tolerance helps to keep the redispatch of active power within reasonable limits. Moreover, due to the computational complexity of this problem, the commitments of the generators cannot be altered. As a result, redispatch only changes the dispatch of market participants, but not their commitment status. The AC power flow problem is thus of a continuous rather than discrete nature. The problem is then solved using commercial software, i.e., interior-point methods that try to find a local solution to the large-scale, non-linear, non-convex problems at hand Nocedal et al. (2009), Wächter and Biegler (2005a,b) and Wächter and Biegler (2006). The obtained allocation (x^{AC}, y^{AC}) is AC-feasible, i.e., $(x^{AC}, y^{AC}) \in \Psi^{AC}$, but it is not AC-optimal, i.e., (x^{AC}, y^{AC}) is not a solution to (1) with Ψ^{AC}.

2.3. Competitive equilibrium theory and pricing

After computing an optimal allocation (x^*, y^*) based on some convex power flow representation, the market operator needs to calculate real power prices $\lambda \in \mathbb{R}^{|N|}$ for electricity at each node.[13] We assume prices of zero for reactive power and only assign non-zero prices to real power. The motivation to allocate and price goods through markets is based on competitive equilibrium theory and has a long history. As we will see, however, the design of v_b and c_s as non-convex valuation or cost functions implies a fundamental problem for pricing.

The promise of markets rests on well-known theoretical results such as the welfare theorems (Arrow and Debreu, 1954). Assuming convex preferences v_b and c_s, demand independence, and perfect competition with divisible goods, there is a set of competitive equilibrium prices that will maximize social welfare. Further, every welfare-maximizing allocation can be supported by a set of equilibrium prices. An equilibrium implies that no market participants wants to deviate from the assigned allocation, and that the outcome is thus envy-free and budget-balanced.

However, many real-world markets exhibit indivisible goods and non-convex preferences. In this context, the concept of a quasilinear utility function was widely adopted to study competitive equilibria (Bikhchandani and Mamer, 1997; Bikhchandani and Ostroy, 2002; Baldwin and Klemperer, 2019). A large part of the literature focuses on Walrasian equilibria, i.e., efficient market outcomes with linear (i.e., item-level) and anonymous prices such that every market participant maximizes their utility. Unfortunately, Walrasian equilibria with indivisible goods generally only exist under very restrictive assumptions on the preference functions (e.g., strong substitutes) (Kelso and Crawford, 1982; Bikhchandani and Mamer, 1997; Baldwin and Klemperer, 2019). Under these assumptions, Walrasian prices can be obtained from the Lagrangian dual of the welfare maximization problem and the welfare theorems can be generalized to markets with indivisibilities (Blumrosen and Nisan, 2007; Bichler et al., 2020). However, when these assumptions do not hold, these properties are generally lost. In general non-convex combinatorial markets, determining the allocation is NP-hard and competitive equilibrium prices need to be non-linear and personalized (Bikhchandani and Ostroy, 2002) or not exist at all (Bichler and Waldherr, 2019).

Electricity markets are a prime example of such markets. In the absence of competitive equilibrium prices, different heuristics have been proposed to serve as alternative pricing rules (Liberopoulos and Andrianesis, 2016). The resulting prices are not unique, but virtually all established pricing rules produce linear and anonymous prices such that they can serve as a baseline for derivatives contracts. Given that Walrasian equilibria are impossible to obtain, these prices compromise on some of the properties thereof. We will discuss these issues along two established pricing rules in electricity markets: Integer Programming (IP) pricing (O'Neill et al., 2005) and Convex Hull (CH) pricing (Hogan and Ring, 2003; Gribik et al., 2007).

To ease our reasoning, we will impose a specific form of valuation and cost functions for the remainder of this paper. Buyers b have concave, piecewise-linear valuation functions v_b based on their real power consumption, represented by a set of bids β_b and upper bounds $q_{b\ell}$. Moreover, for both real and reactive power, they possess an inelastic demand $\underline{P}_b$, $\underline{Q}_b$ and a maximum consumption of $\overline{P}_b$, $\overline{Q}_b$, respectively.

$$
\begin{aligned}
v_b(x_b) = \quad & \max_{x_b, x_{b\ell}} \sum_{\ell \in \beta_b} v_{b\ell} x_{b\ell} \\
\text{s.t.} \quad & \sum_{\ell \in \beta_b} x_{b\ell} = \mathrm{Re}(x_b) \\
& \underline{P}_b \le \mathrm{Re}(x_b) \le \overline{P}_b \\
& \underline{Q}_b \le \mathrm{Im}(x_b) \le \overline{Q}_b \\
& 0 \le x_{b\ell} \le q_{b\ell} \quad \forall \ell \in \beta_b
\end{aligned}
$$

Buyers are thus assumed to have fully convex preferences. On the other hand, for sellers s we define a binary commitment variable u_s that signals if a generator is supplying a positive amount and thereby introduces non-convexities. When $u_s = 1$, seller s has a convex, piecewise-linear variable cost function in analogy to buyers' valuation functions. Moreover, fixed costs h_s occur when a positive amount is produced, and sellers have minimum and maximum production quantities $\underline{P}_s$, $\underline{Q}_s$, $\overline{P}_s$, and $\overline{Q}_s$, respectively.

$$
\begin{aligned}
c_s(y_s) = \quad & \max_{y_s, y_{s\ell}, u_s} \sum_{\ell \in \beta_s} v_{s\ell} y_{s\ell} + h_s u_s \\
\text{s.t.} \quad & \sum_{\ell \in \beta_s} y_{s\ell} = \mathrm{Re}(y_s) \\
& \underline{P}_s u_s \le \mathrm{Re}(y_s) \le \overline{P}_s u_s
\end{aligned}
$$

[12] Note that redispatch also refers to the process of adjusting a dispatch in zonal markets due to the neglect of intra-zonal transmission constraints. In this paper, by redispatch we refer to the process of reconstructing an AC-feasible solution from a solution obtained from a relaxation/approximation.

[13] The prices obtained from the convexified allocation problem also apply to the allocation after redispatch, i.e., for (x^{AC}, y^{AC}).

$$\underline{Q}_s u_s \le \mathrm{Im}(y_s) \le \overline{Q}_s u_s$$
$$0 \le y_{s\ell} \le q_{s\ell} \quad \forall \ell \in \beta_s$$
$$u_s \in \{0,1\}$$

As discussed, with such non-convex cost functions, Walrasian equilibrium prices can generally not be obtained. As such, some market participants do not maximize their utility at the prices λ. We call this forgone payoff *lost opportunity costs* (LOCs) of a market participant, i.e.,

$$LOC_b = \hat{u}_b(\lambda) - u_b(x_b^*|\lambda) = (\max_{x_b} v_b(x_b) - \lambda^T \mathrm{Re}(x_b)) - (v_b(x_b^*) - \lambda^T \mathrm{Re}(x_b^*)),$$

$$LOC_s = \hat{u}_s(\lambda) - u_s(y_s^*|\lambda) = (\max_{y_s} \lambda^T \mathrm{Re}(y_s) - c_s(y_s)) - (\lambda^T \mathrm{Re}(y_s^*) - c_s(y_s^*)).$$

We call $\hat{u}_b(\lambda)$, $\hat{u}_s(\lambda)$ the *indirect utility* function that measures the individual payoff maximum given the prices λ. In contrast, the *direct utility* $u_b(x_b^*|\lambda)$, $u_s(y_s^*|\lambda)$ measures the actual payoff given the optimal allocation (x^*, y^*) and prices λ.

A natural choice for a pricing rule in non-convex markets is to minimize these lost opportunity costs. This is the main idea of Convex Hull (CH) pricing. CH pricing replaces each non-convex cost function $c_s(y_s)$ by its convex envelope $\overline{\mathrm{conv}}(c_s(y_s))$. Assuming a convex power flow representation, prices can be obtained from the dual of the resulting convex problem. Determining convex envelopes and thus calculating CH prices is generally computationally hard (Schiro et al., 2016). However, under restricted bidding languages, as applicable to our form of c_s, CH prices can be tractably obtained by relaxing the binary constraints $u_s \in \{0,1\}$ to $u_s \in [0,1]$ and solving the dual problem (Hua and Baldick, 2017). The resulting prices λ^{CH} then equal the dual variables of the active power balance constraints. They minimize LOCs, and if a Walrasian equilibrium is attainable, λ^{CH} constitutes such Walrasian prices.

Since CH prices cannot always be computed efficiently, most market operators in practice implement Integer Programming (IP) pricing. This involves (i) obtaining the optimal commitment variables u_s^* from the welfare-maximizing solution, (ii) fixing the commitment variables to these optimal values, i.e. relaxing $u_s \in \{0,1\}$ to $u_s \in [0,1]$, $u_s = u_s^*$, and (iii) retrieving prices from the dual of the active power balance constraints of the resulting convex problem. IP pricing is considered to follow the notion of marginal pricing in non-convex markets (O'Neill et al., 2005).

Recently, practical pricing rules that are based on IP pricing have come under scrutiny. They have been criticized for invoking too high *make-whole payments* (MWP) to some of the market participants, distorting the market price signal (Bichler et al., 2023; O'Neill et al., 2019). Again, this criticism refers back to the absence of Walrasian prices. In particular, in non-convex markets with linear and anonymous prices, there may be no price vector that allows all market participants to break even (Bichler et al., 2023). We call this property of non-negative payoffs *individual rationality*. If this property is violated for a market participant, they need to be compensated by an individual side-payment. This MWP amounts to

$$MWP_b = \max(-u_b(x_b^*|\lambda), 0) = \max(-(v_b(x_b^*) - \lambda^T \mathrm{Re}(x_b^*)), 0),$$

$$MWP_s = \max(-u_s(y_s^*|\lambda), 0) = \max(-(\lambda^T \mathrm{Re}(y_s^*) - c_s(y_s^*)), 0).$$

This observation has motivated the development of pricing rules with reduced levels of MWPs (Bichler et al., 2023; O'Neill et al., 2019; Ahunbay et al., 2022). Note that while $MWP_b \le LOC_b$ (and for sellers alike), the objectives to minimize MWPs and LOCs can be conflicting (Ahunbay et al., 2022). It is unclear how CH prices, IP prices, as well as the LOCs and MWPs they imply behave depending on the choice of the power flow representation Ψ.

In addition, we aim to benchmark the quality of the congestion signals provided by the different pricing rules. In particular, congestion on network elements should be reflected in the prices, and if no congestion occurs, prices should be identical throughout the network. Flawed congestion signals have an impact on congestion income distribution, financial hedging, and investment decisions, and are thus of essential importance in practical electricity markets. In order to measure the accuracy of congestion signals, we rely on a metric known as *financial transmission right* (FTR) *uplift* or *potential congestion revenue shortfall* (PCRS) (Garcia et al., 2020). This metric represents lost opportunity costs for the transmission operators. It constitutes the difference between the maximum possible FTR payoffs (respecting the set of feasible power flows Ψ) and the actual congestion income given the optimal allocation.

Let us denote $x_N \in \mathbb{C}^{|N|}$ as the vector that holds the aggregate consumption at each node $i \in N$ as components, i.e. $x_N = (\sum_{b\in B} x_{ib})_{i\in N}$. Analogously, we define $y_N \in \mathbb{C}^{|N|}$ as the aggregate supply at each node, i.e. $y_N = (\sum_{s\in S} y_{is})_{i\in N}$. Then, the PCRS is defined as

$$PCRS = \max_{x,y\in\Psi} \lambda^T(\mathrm{Re}(x_N) - \mathrm{Re}(y_N)) - \lambda^T(\mathrm{Re}(x_N^*) - \mathrm{Re}(y_N^*)).$$

FTRs are allocated based on some set of feasible flows which may not correspond to the actual flows. The notion of the PCRS metric is that prices should be set in a way that the congestion revenues can cover payoffs from a *worst-case* FTR allocation (Garcia et al., 2020). If this were not the case, the resulting shortfall has to be distributed among market participants, which is not trivial (Hogan, 2013). Similar to LOCs and MWPs, the behavior of PCRS under different power flow representations Ψ is unclear and one focus of this study.

3. Data and processing

We apply the different power flow relaxations and pricing rules on the network and bid data provided for the ARPA-E Grid Optimization Competition Challenge II.[14] This competition seeks the development of modern and scalable optimization techniques for solving complex power flow problems. To that end, they provide large-scale and realistic test sets of single-period power flow problems. We consider the power flow models and valuation/cost functions as described in the previous section (for which ARPA-E provides all necessary data), including thermal line limits, and therefore abstract from some of the available data (e.g., switched shunts, transformers, contingencies). In the Appendix, we provide an overview of the notation.

We consider a set of five synthetic networks and 20 scenarios. The problem sizes range from 617 nodes, 94 generators, and 404 consumers to 3970 nodes, 391 generators, and 2744 consumers. This matches the size of realistic power networks, e.g., those of Germany (roughly 438 nodes (DIW Berlin, 2022)) or the United Kingdom (roughly 300 nodes (National Grid, 2022)). The dataset includes even larger networks (up to 31,777 nodes), yet solving tighter non-linear convex relaxations beyond DCOPF turned out to be intractable for these problem sizes. For example, a nodal model of Continental Europe and Ireland would encompass roughly 25,000 nodes and 25,000 generators (ENTSO-E, 2022), and the Californian market consists of roughly 9700 nodes (California ISO, 2018). Note that we ran our experiments on an Intel Xeon E312xx 2.0 GHz machine with 20 cores. European-scale models might be tractable on a super-computer, but scalability is not the focus of our research. We instead aim for an estimate of the impact of different relaxations on prices, MWPs, LOCs, and PCRS. In terms of external validity, our problem instances are large enough and representative of those in the field.

We implemented the models in the Julia programming language. For the different power flow models, we worked with the PowerModels library,[15] which offers readily available implementations of all common power flow representations. We built on this library to enrich the

[14] See https://gocompetition.energy.gov/sites/default/files/Challenge2_Problem_Formulation_20210531.pdf for the problem statement.

[15] See https://lanl-ansi.github.io/PowerModels.jl for the package documentation.

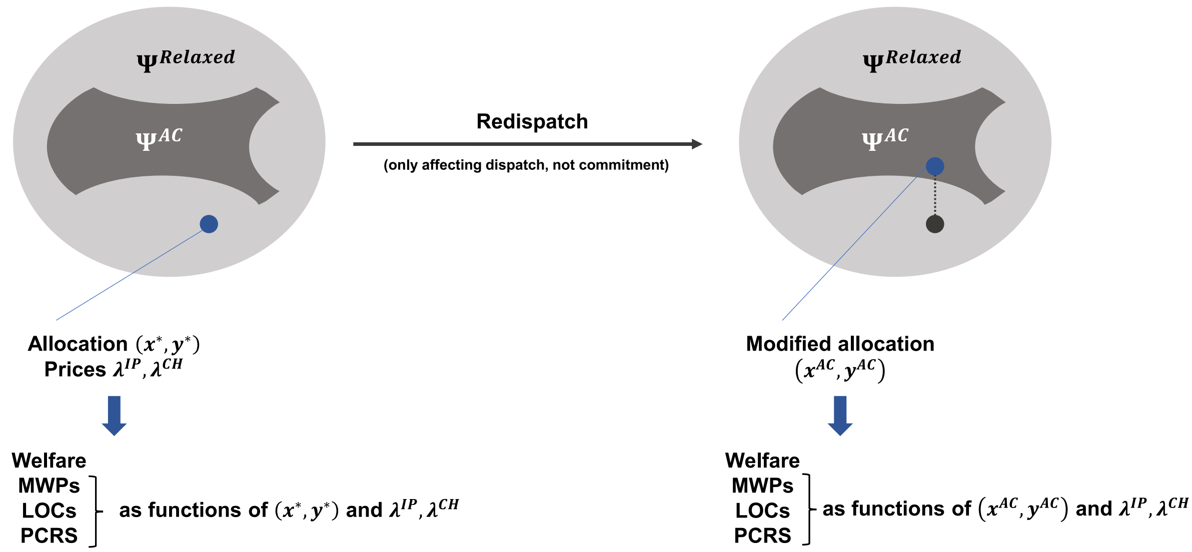

Fig. 1. Visualization of the approach.

Table 1
Overview of experiments.

Network sizes	Power flow models	Pricing rules	Metrics[a]
617 (5 scenarios)	DC	IP	Welfare
2020 (3 scenarios)	SOC	CH	Prices
2312 (5 scenarios)	QC		MWPs
3288 (5 scenarios)			LOCs
3970 (2 scenarios)			PCRS
			Redispatch costs
			Runtime

[a] Welfare, MWPs, LOCs, and PCRS can be analyzed before and after the redispatch is conducted.

models with the non-convex valuation and cost functions described above, and further added functionalities to determine prices, LOCs, MWPs, and PCRS. We used different commercial packages to solve the optimization problems, including Gurobi (DC, SOC, QC), Ipopt (DC, SOC, QC, AC), and Mosek (DC, SOC, QC). With standard constraint qualification conditions satisfied, we can conclude strong duality (Cao et al., 2022) and obtain prices from the dual solutions of our pricing problems. Table 1 summarizes the factors of our experimental design and the metrics analyzed.

As illustrated in Fig. 1, we first compute the allocation and prices for each convexified model. This yields a certain welfare, MWPs, LOCs, and PCRS. Since the computed allocation may not be AC-feasible, we then conduct cost-based redispatch to obtain a physically AC-feasible outcome. The now modified allocation yields a different welfare and – together with the initially computed prices – different MWPs, LOCs, and PCRS. In the next section, we only report MWPs, LOCs, and PCRS for the AC-feasible solution, since this is the outcome that is physically implemented.

4. Results

Below, we report the main results from our analysis.

4.1. Welfare & prices

For each test case, we solve the welfare-maximization problem for each power flow model. This provides the optimal generator dispatch, buyer consumption, and power flows that are valid under the chosen relaxation/approximation.

Result 1. *While the welfare of different power flow models is similar, generator commitments between different power flow models vary significantly. These differences in allocation impact the prices, the level of required redispatch, and the final AC-feasible outcome.*

In terms of welfare, each relaxation/approximation yields a similar outcome. In all tested scenarios, choosing a different power flow model impacts the welfare by, at most, 0.5%. This hinges on the fact that the valuations of the buyers in the data set are very high. Large welfare losses are thus not to be expected under any power flow model, even though minor welfare improvements can already create significant savings (Cain and O'Neill, 2012). At the same time, this does not imply that the allocations are similar for all power flow models. Fig. 2 illustrates the differences in committed generation capacities between Ψ^{DC}, Ψ^{SOC}, and Ψ^{QC}. In the Appendix, we include a similar figure that visualizes differences in the dispatched capacities. In most cases, the conic relaxations commit and dispatch more generators compared to the DC approximation, in order to account for the additional network constraints (e.g., reactive power constraints). The differences can be significant and impact prices and redispatch, as we will see.

Result 2. *While the average prices of different power flow relaxations are similar, prices under Ψ^{DC} exhibit significantly more outliers. The DC solution may contain line congestions that do not occur under Ψ^{SOC} and Ψ^{QC}, leading to potentially high and unjustified price peaks in the network. In contrast, prices for Ψ^{SOC} and Ψ^{QC} are very similar, more stable, and better reflect the physics of the network. CH prices usually exhibit fewer discrepancies between power models than IP prices, but this comes at the expense of distortions in the congestion signals.*

After computing the welfare-maximizing allocation based on the respective power flow representations, we determine IP and CH prices by solving a convexified version of the original allocation problem. We present exemplary boxplots of the computed prices for the networks with 2020 and 3970 nodes (aggregated over all scenarios) in Fig. 3. Especially in large networks, IP and CH prices under Ψ^{DC} exhibit substantially more outliers. We provide small examples that illustrate how such deviations can occur with DCOPF in the Appendix. For example, the very high DC prices up to \$743/MWh for test case C2FEN02020/121 are caused by a single congested line in the respective part of the network, which implies high marginal costs and leads to high prices at all surrounding nodes. Importantly, this line congestion vanishes under Ψ^{SOC} and Ψ^{QC} and prices range at stable levels at around \$75/MWh. While we cannot compute the AC-optimal

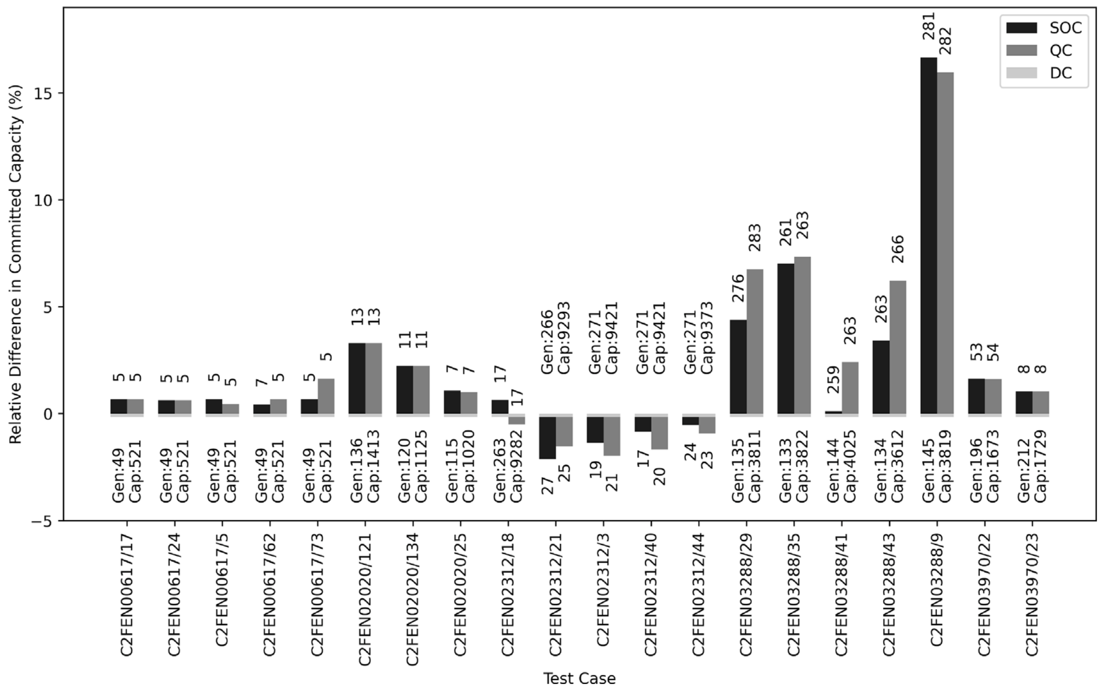

Fig. 2. Committed generation capacity: The figure denotes the number of committed generators ("Gen") and their total capacity ("Cap") for Ψ^{DC} (in MW) at the bottom. The bars indicate the relative difference (in %) in committed capacity for Ψ^{SOC} and Ψ^{QC}, respectively. The labels of each bar indicate the number of generators with a different commitment status compared to Ψ^{DC}. For example, the first bar in the first test case C2FEN00617/17 indicates that 5 units are committed differently under Ψ^{SOC} as compared to the 49 committed units under Ψ^{DC}.

Table 2
IP price statistics.

	μ_{DC}	σ_{DC}	μ_{SOC}	σ_{SOC}	μ_{QC}	σ_{QC}	$\rho_{DC,SOC}$	$\rho_{DC,QC}$	$\rho_{SOC,QC}$
617	112.36	0.00	116.11	2.15	115.29	2.18	0.73	0.73	1.00
2020	76.63	24.86	71.63	3.60	72.16	3.63	0.49	0.49	1.00
2312	9.27	2.44	9.38	2.04	9.40	1.99	0.80	0.79	0.94
3288	18.71	3.04	17.48	0.19	17.46	0.19	0.11	0.14	0.98
3970	28.92	3.72	27.92	0.29	27.92	0.29	0.23	0.23	1.00

Table 3
CH price statistics.

	μ_{DC}	σ_{DC}	μ_{SOC}	σ_{SOC}	μ_{QC}	σ_{QC}	$\rho_{DC,SOC}$	$\rho_{DC,QC}$	$\rho_{SOC,QC}$
617	112.87	0.12	115.00	2.07	115.00	2.06	0.21	0.21	1.00
2020	74.27	23.34	74.58	4.44	74.58	4.44	0.56	0.56	1.00
2312	9.28	2.38	9.34	2.05	9.34	2.05	0.82	0.82	1.00
3288	17.94	2.37	17.59	0.19	17.59	0.19	0.20	0.20	1.00
3970	28.96	3.43	27.94	0.19	27.94	0.29	0.24	0.24	1.00

Table 4
Redispatch costs [$] and their proportion of welfare [%].

	DC	SOC	QC
617	14 709.70 (1.93%)	7452.95 (0.98%)	8269.10 (1.08%)
2020	70 468.60 (2.77%)	42 905.50 (1.73%)	43 580.50 (1.75%)
2312	20 624.10 (0.69%)	390.50 (0.01%)	376.75 (0.01%)
3288	48 896.30 (0.91%)	16 993.70 (0.32%)	12 461.90 (0.23%)
3970	1489.50 (1.25%)	998.41 (0.80%)	992.46 (0.83%)

allocation in this case, we argue that the tighter SOC and QC relaxations better reflect the actual physical flows in the network. Moreover, when computing a near AC-feasible solution, the line congestion vanishes even for the DC outcome. We therefore argue that the high price peaks of the DC network are unjustified and that the conic relaxations provide a better price signal.

The price peaks for Ψ^{DC} lead to a higher volatility (σ) of DC prices, as evident from Tables 2 and 3. While average prices (μ) are similar for different power flow relaxations, DC prices exhibit very little correlation (ρ) with SOC or QC prices. Correlations tend to be higher for CH prices compared to IP prices, with the exception of the 617-node network where DC prices tend to be constant throughout the network.

This is further illustrated by the median prices for each tested scenario in Fig. 4. They are depicted in terms of their relative deviation to the median IP and CH price under Ψ^{DC}, respectively. Generally, price spreads between different power flow relaxations tend to be higher for IP pricing. This can be explained by the design of the convexified model underlying IP and CH pricing. In particular, IP pricing fixes the commitment variables u_s of each generator to their optimal values u_s^* and computes marginal prices based on this assumption. In contrast, CH pricing computes marginal prices based on the convex envelopes of the non-convex cost functions. As a result, generators under the convexified model for CH pricing can be dispatched more flexibly than under the convexified model for IP pricing. Differences in generator commitments thus do not impact CH prices, and hence CH prices are more similar across different power flow representations. Notably, in all tested scenarios the conic relaxations do not increase prices significantly compared to the DC benchmark, and often even decrease median prices by several percent.

4.2. Redispatch costs and AC-feasibility

Next, we consider cost-based redispatch to obtain an AC-feasible solution according to the methodology described in Section 2.2. Note that redispatch refers to the allocation alone, and the choice of pricing rule therefore does not matter. Redispatch only changes the dispatch quantities of buyers and sellers, but not their commitment status.

Result 3. *Redispatch costs are substantial for the DC approximation. The required adjustments of the allocation to obtain AC-feasibility correlate with the tightness of the power flow model. The high redispatch costs of the linear DC approximation can be significantly reduced by using second-order conic power flow relaxations.*

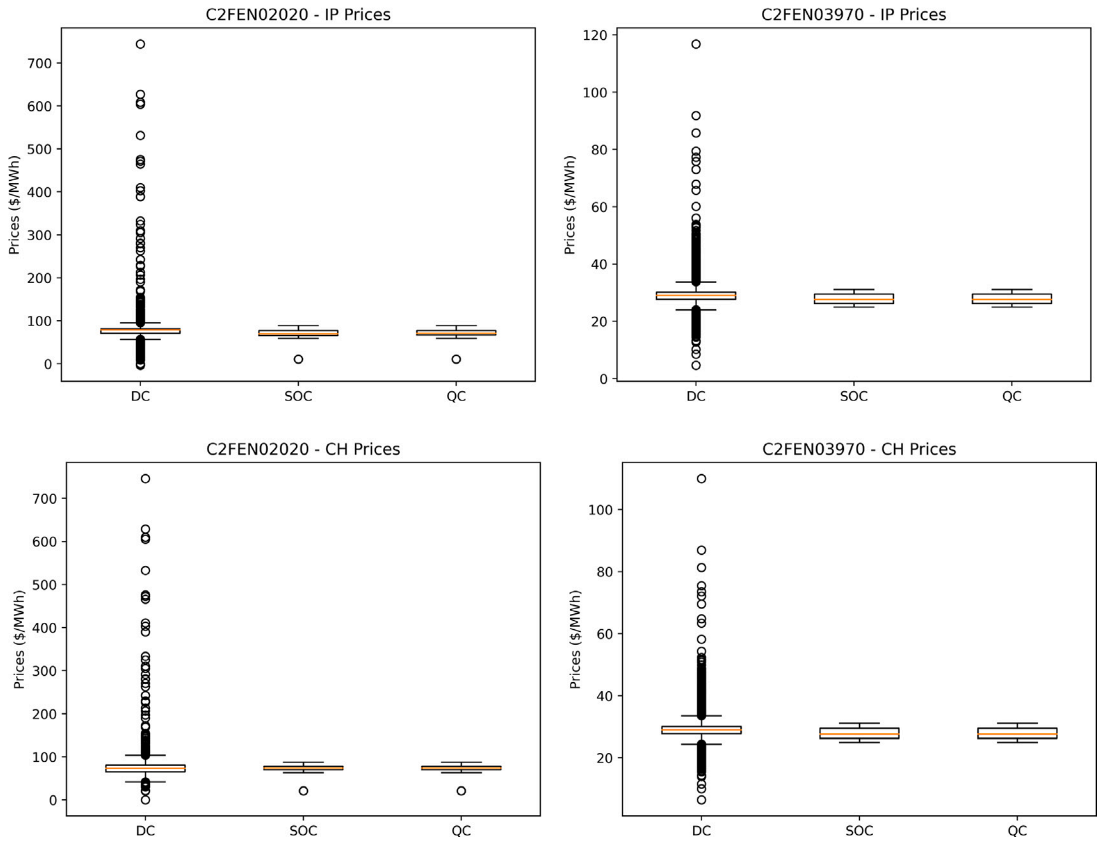

Fig. 3. Price boxplots [$/MWh].

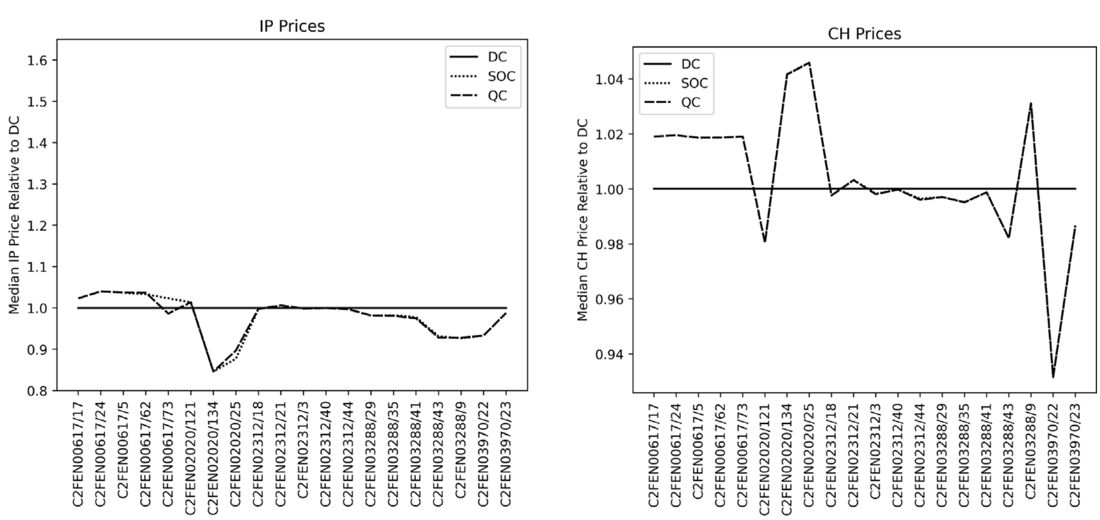

Fig. 4. Median prices.

Table 5
Average welfare, MWPs, LOCs, and PCRS for the AC-feasible solutions.

	DC		SOC		QC	
	IP	CH	IP	CH	IP	CH
Welfare vs. Ψ^{DC}_{AC}						
617	–	–	+0.91%	+0.91%	+0.83%	+0.83%
2020	–	–	+1.10%	+1.10%	+1.09%	+1.09%
2312	–	–	+0.72%	+0.72%	+0.73%	+0.73%
3288	–	–	+0.40%	+0.40%	+0.50%	+0.50%
3970	–	–	+0.53%	+0.53%	+0.53%	+0.53%
MWPs						
617	1.55	0.00	0.00	0.00	44.39	0.64
2020	5375.92	504.37	1927.20	400.76	1789.93	400.76
2312	51 766.40	50 824.40	51 029.00	50 302.70	50 750.70	50 121.60
3288	36.09	7.54	9536.83	9527.38	9624.19	9611.17
3970	49.53	24.87	16.94	16.93	16.12	16.12
LOCs						
617	13 801.00	13 779.10	7187.81	7041.99	7682.68	7536.17
2020	75 277.60	68 553.00	43 440.30	41 936.50	43 361.20	41 960.10
2312	91 967.90	90 953.10	72 330.60	71 160.00	72 648.50	70 975.40
3288	47 815.60	47 309.10	25 898.20	25 887.50	21 538.00	21 522.10
3970	692.68	672.05	457.66	457.70	456.49	456.55
PCRS						
617	1106.21	1109.01	169.16	200.15	268.47	298.14
2020	2437.87	2245.06	899.50	1042.09	1008.50	1054.98
2312	425.05	450.97	49.34	77.67	76.53	247.45
3288	1023.04	988.94	401.89	415.09	167.15	182.74
3970	526.27	523.25	112.99	114.19	112.99	114.20

Table 4 lists the redispatch costs for all tested scenarios. As expected, the DC approximation requires the highest degree of redispatch to obtain an AC-feasible outcome. On average, the redispatch costs for the DC approximation exceed those of the SOC relaxation by a factor of 2.49, and those of the QC relaxation by a factor of 2.67. The linearized DCOPF abstracts too much from physical reality, and large additional costs arise from correcting the dispatch. Even if these costs are not passed on to consumers,[16] the significant change in allocation implies that the prices do not reflect accurate scarcity signals. In contrast, the two conic relaxations are tighter and thus approximate Ψ^{AC} much better. As a result, on average, significantly less redispatch is necessary.

Result 4. *As redispatch modifies the underlying allocation, the originally computed prices lead to a different outcome. Considering final AC-feasible solutions, tighter convex power flow relaxations imply higher welfare and fewer incentives to deviate, as measured by MWPs, LOCs, and PCRS. The price signals obtained from the SOC or QC relaxation are much more suitable for the AC-feasible outcomes than standard DC prices. This results from the fact that the SOC and QC relaxations are better representations of the physical grid than the DC approximation.*

After conducting the redispatch, we have obtained an AC-feasible dispatch that satisfies grid constraints and can be physically realized. Table 5 summarizes the final welfare, MWPs, LOCs, and PCRS as a result of redispatch.

As discussed above, the redispatch is most substantial for the DC approximation, resulting in the highest welfare losses from ensuring AC-feasibility. In Table 5, we compare the welfare of the AC-feasible outcomes of the conic relaxations relative to the welfare of the AC-feasible outcome of the DC approximation (denoted Ψ^{DC}_{AC}). The welfare based on the conic relaxations outweighs that of the DC approximation in all tested scenarios. Thus, even though we cannot compute the welfare-maximizing AC-optimal outcome, tighter non-linear relaxations provide an allocation that enables higher welfare than the standard linearized DC approximation. Using estimates from the past, these efficiency gains between 0.40% and 1.10% could translate into billions of dollars in annual savings (Cain and O'Neill, 2012), especially with the recent increase in energy prices.

Importantly, while the originally computed prices are not modified, the underlying allocation changes as a result of the redispatch and implies certain MWPs and LOCs for market participants. The highest MWPs are necessary for IP pricing and Ψ^{DC} in 14 of the 20 tested scenarios. With CH pricing the MWPs are consistently lower, yet the DC approximation again requires the highest MWPs in the majority of scenarios when CH pricing is applied. One exception is the 3288 node network, where the set of generators that require MWPs under Ψ^{DC} and under Ψ^{SOC}/Ψ^{QC} are entirely disjunct. However, even though the tighter conic relaxations require higher MWPs, their welfare exceeds that of Ψ^{DC} even more (e.g., by an average of $26,611 for the QC relaxation).

Similarly, we observe a strong increase in LOCs for the DC approximation as a result of the redispatch. In particular, LOCs are highest for Ψ^{DC} in 19 of 20 cases. We thus conclude that the redispatch of the DC approximation can lead to substantial changes in the allocation (e.g., for reactive power balancing). As a result, prices based on Ψ^{DC} tend to be more distorted for the final AC-feasible outcome than prices based on tighter convex relaxations. They imply strong incentives for market participants to deviate from the AC-feasible dispatch. Large amounts of penalties are necessary to prevent market participants to deviate. In contrast, price signals obtained from non-linear relaxations maintain better incentives for the final AC-feasible dispatch.

Moreover, PCRS increase over-proportionally for Ψ^{DC}, implying that prices based on Ψ^{DC} do not reflect congestion properly and can imply significant PCRS for transmission operators. In contrast, the PCRS for the SOC and QC relaxation are significantly smaller, and the respective prices admit a much higher quality in terms of congestion signals for the final AC-feasible dispatch than the DC prices.

4.3. Computational costs

Finally, we consider the computational costs to determine allocation and prices.

[16] In decentralized European electricity markets, redispatch costs are passed on to consumers. In U.S. central dispatch markets, the independent system operator can mandate an AC-feasible dispatch as the final allocation.

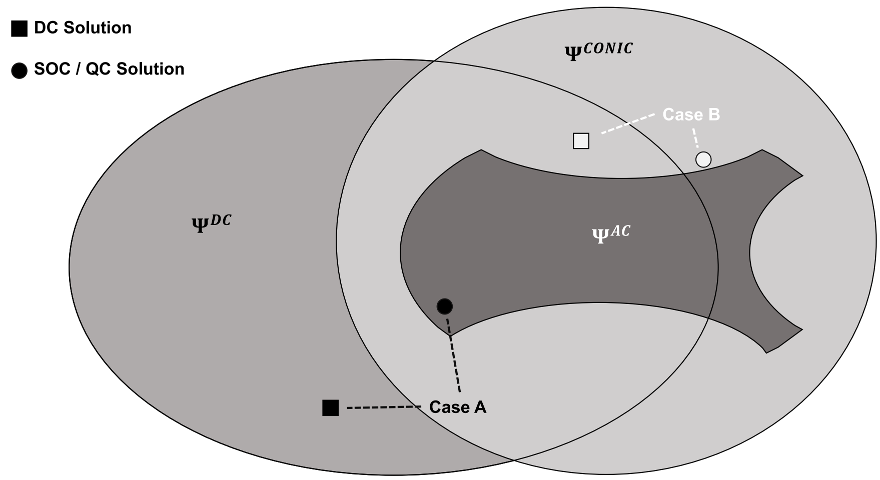

Fig. 5. Visualization of the examples.

Result 5. *Despite advances in non-linear optimization, computational effort is still a limiting factor when employing tighter power flow relaxations. We could solve all mixed-integer second-order conic and quadratic convex relaxations to optimality for the selected problem sizes. Semidefinite programs, however, could not be solved on our test installation as mixed-integer problems, and could not reliably be solved as continuous problems for larger problem sizes.*

When computing the original allocation, mixed-integer programs have to be solved in order to determine generator commitments. While such problems are NP-hard in general, the mixed-integer linear programs underlying the DC approximation can be solved effectively by modern solver technology for the problem sizes at hand. On our machine, the median runtime over all scenarios is only 2.22 s (maximum 12.16 s). Although second-order cone program solvers have matured, solving the mixed-integer SOCPs means more computational effort. As such, the median runtime for the SOC relaxation is 9.1 min (maximum 382.4 min) and for the QC relaxation it is 11.4 min (maximum 564.4 min).

Naturally, the runtimes for the continuous-type pricing problems decrease compared to the respective mixed-integer problems. In particular, the runtimes for the second-order cone programs decrease significantly to a median runtime of 81.5 s (307.4 s) over all pricing problems for the SOC (QC) relaxation, respectively. We note that we only consider a single-period problem, and that more pronounced runtime discrepancies are to be expected in a multi-period context with intertemporal constraints (e.g., ramping constraints).

5. Conclusion

Clearing electricity spot markets is a complex problem. A feasible dispatch has to consider non-linear transmission constraints and non-convex preferences of the market participants. As a result, the ACOPF clearing problem is intractable to solve and market operators need to rely on convex relaxations or approximations of network constraints. Besides, buyers and sellers typically possess non-convex preference functions, preventing the existence of Walrasian equilibrium prices. As a consequence, market participants can have incentives to deviate or even incur losses that need to be compensated by individual side-payments. In this paper, we aim to bring together the literature of optimal power flow and competitive equilibrium theory and study clearing and pricing in non-convex electricity markets with non-linear network models.

Crucially, prices based on the standard linearized DC approximation can be fundamentally flawed and insufficiently reflect actual network flows. Prices can wrongly signal congestion that does not comply with actual scarcities in the AC-feasible outcomes. This is important since consumers may be exposed to unjustifiably high prices, and investment and demand response signals as well as congestion rents are fundamentally distorted.

One reason for diverging prices is the fact that the DC approximation does not price reactive power, even though it affects the final AC-feasible allocation. Given these findings, one direction for future research could be an assessment of whether reactive power prices on the transmission level might be a reasonable addition for a future market design. Reactive power sources are currently compensated as ancillary services (Yu et al., 2019), but if reactive power constraints are binding for the allocation, a locational reactive price signal might better reflect the physical reality.

Moreover, our experiments reveal a number of additional insights. The allocations and committed generation capacities can differ significantly between different power flow models. This has an impact on the properties of prices and the required redispatch to enable AC-feasibility. Typically, more capacity is committed and dispatched with tighter convex relaxations, as this is required to accommodate the more complex transmission constraints.

In terms of pricing rules, CH prices are less dependent on the underlying power flow model. They exhibit less MWPs and minimize LOCs of market participants. CH prices are generally intractable to compute and the prices might not reflect congestion accurately, resulting in larger PCRS for the transmission operator. In contrast, IP prices provide better congestion signals and lower PCRS. However, they are more volatile and depend on the chosen power flow model.

The solution obtained from a DC approximation is often far away from an AC-feasible outcome, requiring significant redispatch. Tighter convex relaxations reduce redispatch costs substantially. Moreover, they lead to higher welfare and much improved price signals. The computational costs for dispatch and pricing remains an advantage for the DC approximation. The underlying (mixed-integer) linear programs scale better than equivalent second-order conic (or even semidefinite) programs. However, the second-order conic relaxations provide a viable alternative already today. An open question for future research is the evaluation of convex power flow relaxations for very large (e.g., European-scale) grids and multi-period problems.

Overall, we conclude that tighter power flow relaxations can provide significant upsides for electricity spot markets. Apart from welfare gains and low redispatch costs, we argue that unbiased price signals are a major advantage of advanced power flow models. This is even more relevant for future electricity markets with many decentralized renewable energy sources, variability in supply and demand, and less predictable congestion patterns.

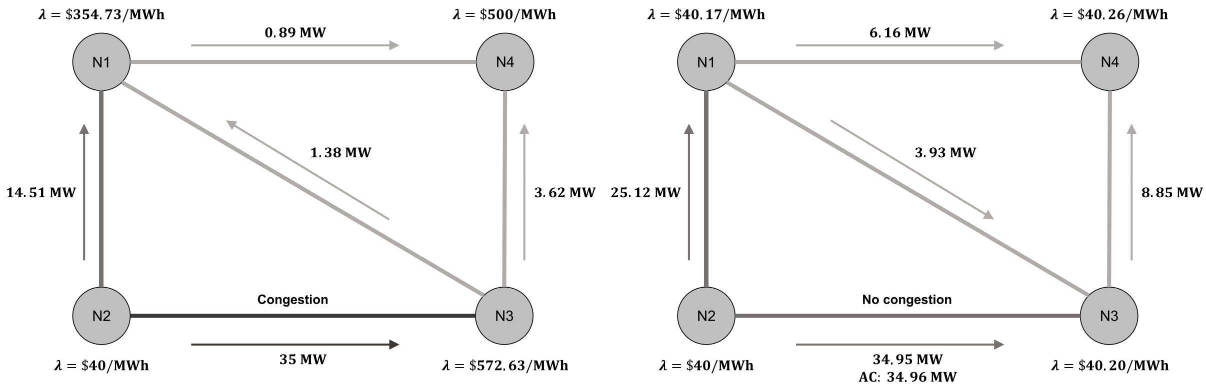

Fig. 6. DC (Left) and SOC (Right) Prices and line flows.

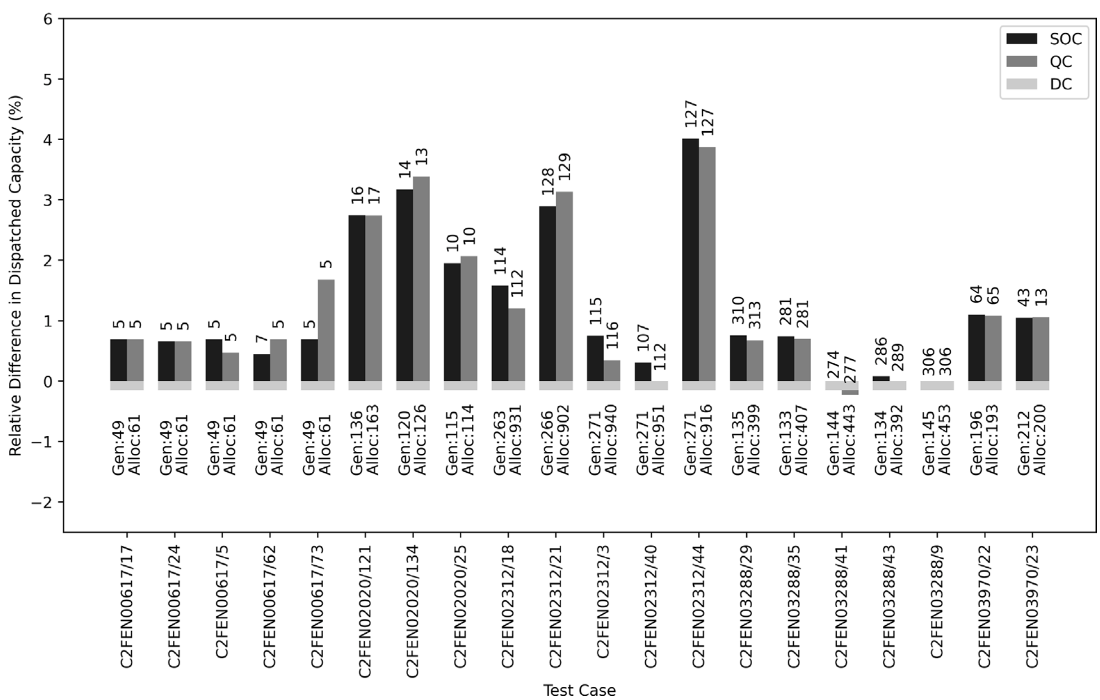

Fig. 7. Dispatched generation capacity: The figure denotes the number of dispatched generators ("Gen") and their total dispatched capacity ("Alloc") for Ψ^{DC} (in MW) at the bottom. The bars indicate the relative difference (in %) in dispatched capacity for Ψ^{SOC} and Ψ^{QC}, respectively. The labels of each bar indicate the number of generators with a different dispatch compared to Ψ^{DC}. For example, the first bar in the first test case C2FEN00617/17 indicates that 5 units were dispatched differently under Ψ^{SOC} as compared to the 49 dispatched units under Ψ^{DC}.

CRediT authorship contribution statement

Martin Bichler: Conceptualization, Validation, Resources, Writing – original draft, Writing – review & editing, Supervision, Funding acquisition. **Johannes Knörr:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Project administration.

Appendix A. Biased prices from the DCOPF approximation

Currently, prices on electricity markets rely on the linearized DC approximation, regardless of the chosen pricing rule (e.g., IP or CH). A main finding of our numerical experiments concerned the large price discrepancies at some nodes that appeared even though there was no congestion in the AC-feasible solution.

Large networks as those analyzed in the paper are difficult to study. In this section, we illustrate how such problems can arise in very small networks with two or four nodes. Even in these small examples, DC prices may signal congestion where there is none, or fail to signal congestion even though it occurs.

Fig. 5 visualizes Ψ^{AC}, Ψ^{DC}, and one exemplary conic relaxation (either Ψ^{SOC} or Ψ^{QC}). Note that Ψ^{SOC} and Ψ^{QC} represent *relaxations*, i.e., $\Psi^{AC} \subseteq \Psi^{SOC}$ and $\Psi^{AC} \subseteq \Psi^{QC}$, while Ψ^{DC} represents an *approximation*, i.e., $\Psi^{AC} \setminus \Psi^{DC}$ might be non-empty. Generally, we are unable to determine the AC-optimal allocation (due to computational hardness) and AC-optimal prices (due to computational hardness and the non-existence of competitive equilibria). For the small examples below, however, we can at least compute the AC-optimal allocation and analyze the quality of different price signals based thereupon.

Case A: DC-optimal solution being SOC/QC-Infeasible

We consider a simple market with two connected nodes N1 and N2 and a single time period. Both nodes have a base voltage of 1 p.u. and a voltage angle of 25 degrees. A single generator at N1 can supply up to 100 MW at a cost of $40/MWh. A single buyer at N2 wishes to consume up to 30 MWh for $3000/MWh. Both market participants

Table 6
Lines.

ID	From-node	To-node	Resistance [p.u.]	Reactance [p.u.]	Thermal limit [MW]
L1	N1	N2	0.01	0.03	35
L2	N1	N3	0.01	0.03	25
L3	N1	N3	0.02	0.04	30
L4	N2	N3	0.005	0.01	35
L5	N3	N4	0.01	0.02	30

Table 7
Mapping of notation.

Notation in this work	Description	Equivalent notation for ARPA-E
Sets		
B	Buyers	J
S	Sellers	G
N	Nodes	I
L	Lines	E and F
Parameters		
V_i^{min}	Minimum voltage at node $i \in N$	$\underline{v}_i$
V_i^{max}	Maximum voltage at node $i \in N$	$\overline{v}_i$
G_{ik}	Conductance at line $(i,k) \in L$	$\mathrm{Re}((R+jX)^{-1})$
B_{ik}	Susceptance at line $(i,k) \in L$	$\mathrm{Im}((R+jX)^{-1})$
$v_{b\ell}$	Valuation for bid $\ell \in \beta_b$ of buyer $b \in B$	c_{jn}
$q_{b\ell}$	Upper bound for bid $\ell \in \beta_b$ of buyer $b \in B$	$\overline{p}_{jn}$
$\underline{P}_b$	Minimum real power of buyer $b \in B$	$p_j^0 \underline{t}_j$
$\overline{P}_b$	Minimum real power of buyer $b \in B$	$p_j^0 \overline{t}_j$
$\underline{Q}_b$	Minimum reactive power of buyer $b \in B$	$q_j^0 \underline{t}_j$
$\overline{Q}_b$	Minimum reactive power of buyer $b \in B$	$q_j^0 \overline{t}_j$
$v_{s\ell}$	Variable cost for bid $\ell \in \beta_s$ of seller $s \in S$	c_{gn}
h_s	Fixed cost of seller $s \in S$	c_g^{on}
$q_{s\ell}$	Upper bound for bid $\ell \in \beta_s$ of seller $s \in S$	$\overline{p}_{gn}$
$\underline{P}_s$	Minimum real power of seller $s \in S$	$\underline{p}_g$
$\overline{P}_s$	Minimum real power of seller $s \in S$	$\overline{p}_g$
$\underline{Q}_s$	Minimum reactive power of seller $s \in S$	$\underline{q}_g$
$\overline{Q}_s$	Minimum reactive power of seller $s \in S$	$\underline{q}_g$
Variables		
x_b	Consumption of buyer $b \in B$	p_j
$x_{b\ell}$	Consumption of buyer $b \in B$ regarding bid $\ell \in \beta_b$	p_{jn}
y_s	Generation of seller $s \in S$	p_g
$y_{s\ell}$	Generation of seller $s \in S$ regarding bid $\ell in \beta_s$	p_{gn}
u_s	Commitment of seller $s \in S$	x_g^{on}
$\lvert V_i \rvert$	Voltage magnitude at node $i \in N$	v_i
θ_i	Voltage angle at node $i \in N$	θ_i

have no preferences or restrictions with respect to reactive power. They both have fully convex preference functions, and thus IP and CH prices are equivalent. The line connecting N1 and N2 has a resistance of 0.02 p.u., a reactance of 0.01 p.u., and a thermal line capacity of 30.1 MW.

The optimal solution based on Ψ^{DC} is to let the generator supply 30 MWh to the buyer. The lossless DC line transmits 30 MWh and is thus uncongested. The price at both nodes is \$40/MWh.

In this example, the SOC and QC solutions are identical. Since these power flow models consider line losses, the DC-optimal solution is not SOC/QC-feasible. In particular, under either conic model, the generator produces 30.1 MWh and transmits this to N2, yet due to line losses, the buyer only obtains 29.95 MWh. The transmission line is congested due to its thermal line limit, and the prices are \$40/MWh at N1 and \$3000/MWh at N2.

Importantly, the SOC/QC-solution is equivalent to the AC-optimal allocation. In other words, the welfare-maximizing physically feasible AC-solution implies a congested transmission line, and a price difference between the two nodes is justified or even necessary. The prices based on Ψ^{DC} significantly distort the congestion income of the transmission operator and send wrong investment or demand response signals. Prices based on Ψ^{SOC} or Ψ^{QC} are much better suited for the actual AC-optimal outcome.

Case B: SOC/QC-optimal solution being DC-Infeasible

We now consider a market with four nodes and five transmission lines. All nodes have a base voltage of 1 p.u. and a voltage angle of 25 degrees, and the line data are shown in Table 6.

A generator at N2 can supply up to 100 MW at a cost of \$40/MWh, and a generator at N4 can provide up to 30 MW at \$500/MWh. At both N1 and N4, a buyer is willing to consume up to 15 MWh for \$3000/MWh. At N3 a buyer can consume up to 30 MWh for \$3000/MWh. Similar to above, there are no restrictions regarding reactive power, IP and CH prices are equivalent due to fully convex market participants, and the SOC and QC relaxations produce the same prices.

Under all allocations, all buyers consume their maximum quantities. However, the implied power flows are different. Under Ψ^{DC}, the thermal capacity of line L4 is fully exhausted and the network is thus congested. This leads to large price differences, ranging from \$40/MWh at N2 to \$572.63/MWh at N3. In contrast, under the optimal SOC and QC allocation, the critical line L4 is not fully exhausted. As a result, the network is uncongested, and prices are roughly \$40/MWh at each location (minor price differences occur due to line losses). This is illustrated in Fig. 6.[17] The power flows of the SOC/QC-relaxation are not DC-feasible.

Importantly, under the AC-optimal outcome, none of the transmission lines are congested, including the critical line L4. We argue that in this case only minor price differences due to line losses should occur in the network, as implied by the conic relaxations. In contrast, the linearized DC approximation sets unjustifiably high prices at some of the nodes, which comes at the expense of electricity consumers at these locations.

While the test cases above are illustrative, we observe the same patterns in Section 4. While we cannot verify the AC-optimal outcome for these large networks, we argue that the price signals obtained from tighter conic relaxations reflect the physical network flows much better, avoiding the highly distorted prices, congestion incomes, or demand response signals that DC prices can imply.

Appendix B. Mapping of notation to ARPA-E documentation

In the following, we provide a mapping of the notation used in this paper to the notation used in the problem formulation for the ARPA-E Grid Optimization Competition Challenge II. We refer to the following document: https://gocompetition.energy.gov/sites/default/files/Challenge2_Problem_Formulation_20210531.pdf. Note that some parameters from ARPA-E are normalized (see Table 7).

Appendix C. Differences in dispatched capacities

Fig. 2 in the main part of this paper illustrates the differences in *committed* capacity during unit commitment. In contrast, the following Fig. 7 illustrates the difference in *dispatched* capacity. In particular, the figure visualizes the differences in allocation across different power flow models. As Fig. 2 already implied, allocations can vary significantly depending on the choice of power flow model, with up to 4.01% differences in dispatched capacities. This, of course, has an impact on pricing and the required redispatch.

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